Fire safety for a family home · early warning and safe exit · Taif / Palm House V7.1

Owner decision 1 Nov 2026 (design review E-08, critical) · the adopted scheme, its verification (§ 5) and the family escape plan (§ 6) · the rules cited; the Saudi 2024 texts are **not verified** here: the designer of record confirms them before the permit set

1 · The strategy, as adopted

- **Early warning:** mains-powered (hardwired) smoke alarms with a built-in sealed 10-year lithium battery back-up, interconnected by wire, working without internet, all from one compatible range; fed from the essentials circuit if the electrical designer agrees (the battery back-up remains essential in a power cut; the generator is not needed for the alarms). A heat alarm of the same range in the kitchen; a gas detector near the hob with an automatic shut-off valve if the gas installer offers it; one CO alarm on the gas / generator side. Maintenance: monthly test, cleaning per the manufacturer, every alarm replaced after ≈ 10 years.
- **Safe exit:** alarm → the stair → one of the three ground-floor exits (the family entrance, the kitchen service door, the guest entrance through the foyer); every exit door opens from inside without a key (thumb-turn). Backup if the open stair fills with smoke: the master suite uses the suite court (open to the sky) as a fresh-air refuge; the other bedrooms open their escape casements for air, call for help and wait for rescue from outside.
- **Escape windows:** the master bedroom, bedrooms 1–3 and the kids' room have hinged glass casements opening fully from inside, with no grilles, bars or fixed mashrabiya (the five screens removed; mashrabiya only on non-escape windows). Child restrictors with an adult release may be added after occupation (note only). The maid room: one casement ≈ 1.14 × 1.00 m, sill +1.10, obscured glass below +1.60, no screen.
- **Living ventilation:** the high windows of the double-height living open outward (top-hung, rain-safe): W3's lower lites and the two south clerestories, worked from the floor by a wall switch with small chain actuators (or a pole): a normal ventilation feature, useful to clear smoke. Bedroom doors: normal solid timber doors.
- **Equipment:** a fire blanket and an extinguisher in the kitchen; one extinguisher on each floor near the stair.

4 · Not adopted (reason: home-level strategy, early warning and safe exit)

sprinklers (NFPA 13D) · fire-rated doors with closers · a central alarm panel · beam / aspirating detectors in the void · a second door from the men's majlis · a safe room (security = CCTV, alarm and video intercom; R-33 updated). The gas cabinet and the generator point come with the site plan.

Summary of the verification (§ 5)

- **Escape windows:** 6 sleeping rooms, **6 pass, 0 fail** with the screens removed and the maid window at +1.10 (table, page 2).
- **Alarms:** 14 smoke alarms + 1 heat alarm + 1 CO alarm + the gas detector, placed on both plans (page 3); one inside every bedroom and the maid room **OK** the ground-floor alarms ≥ 3 m from the hob **OK** all ≥ 0.9 m from the shower bathrooms' doors **OK** audibility: an alarm inside each bedroom meets the 75 dBA-at-the-pillow practice, to be tested on site (**not verified** until tested).
- **Kitchen:** the dining door is a solid door; the kitchen walls are 200 mm block to the slab; but the **kitchen ↔ laundry entrance has no door**, so smoke reaches the laundry and the maid room's door: **problem** → proposal P1 (page 4).
- **Maid window:** net 1.04 × 0.9 = 0.93 m², one movement, no screen, the strip outside open at both ends: **OK**.
- **Bedroom windows and rescue from outside:** all pass; the west strip is 0.785 m wide and a ladder needs the driveway beyond the M line: **not verified** → note for the site plan (page 2).
- **Exit doors:** the family entrance 0.90 m clear (egress door **OK**); the service door 0.80 and the guest double door 1.30 clear with both leaves (secondary exits); thumb-turns on all three: a hardware specification (**OK** once specified).
- **Second way out of the first floor (south wing v2, 1 Nov 2026): house → suite court → office → external stair → outside: NOT COUNTED as designed today** — from the stair top along the gallery to the gym's sliding door, the gym's court door (into the court, free from the gym side), the 1.2 m strip along the court's north edge, the office's court door (into the office, free from the court side, key only from the office side), the visitors' door (out onto the landing, thumb-turn inside), down two flights to the foot gate (outwards, panic bar): 24.2 m from the stair top, 8.2 m from the suite bedroom. **Doors and gates:** all open in the escape direction or slide, none needs a key on the way out (table, page 4). **The court's exposure:** open to the sky (smoke rises away), its west, east and south sides solid or open to the garden; but its **north side is the double-height living's south wall**: the two clerestory windows (+5.00 … +6.00, 1.40 m above the court floor) are ordinary glass and **openable vents for smoke clearing**, directly above the strip; a fire on the ground floor (the living, the kitchen, the dining: the case in which the internal stair is lost) vents smoke and, after the glass fails, flames into the court over the only route. **The reason it is not counted: the clerestory windows. Condition to count it:** make them fixed fire-resisting

The rules checked against

Rule	Source	Status
Escape and rescue opening in every sleeping room: net clear ≥ 0.53 m² (0.47 at grade), ≥ 0.61 high × 0.51 wide, sill ≤ 1.12 operable from inside without keys, tools or special knowledge; grilles and screens only if releasable from inside	IRC 2021 R310.1–R310.4 · SBC 1101 R310	not verified
One egress door ≥ 0.81 m clear, openable from inside without a key; egress from every part of the dwelling	IRC 2021 R311.1–R311.2 · SBC 1101 R311	not verified
Smoke alarms in each sleeping room, outside each sleeping area, on every storey; interconnected; mains with battery back-up (UL 217); ≥ 3 m from a cooking appliance (photoelectric), ≥ 0.9 m from a bathroom door with a shower	IRC 2021 R314.3–R314.6 · NFPA 72-2022 29.8.3, 29.11 · SBC 1101 R314	not verified
Ceiling-mounted alarms ≥ 0.1 m from any wall; wall-mounted 0.10–0.30 m below the ceiling; not in dead air	NFPA 72-2022 29.11.3 · manufacturer's instructions	not verified
Alarm audibility ≥ 75 dBA at the pillow in every sleeping room (an alarm inside the room meets it)	NFPA 72-2022 29.5.2 (practice; the IRC asks only for interconnection)	not verified
Carbon-monoxide alarm outside the sleeping areas where a fuel-fired appliance is installed (the gas hob)	IRC 2021 R315 · SBC 1101 R315	not verified
LP-gas cylinders outside in a ventilated cabinet; extinguishers; a gas shut-off	SBC 801 (from the IFC) · Civil Defense villa rules	not verified

Not verified = the figure was read in the IRC / NFPA text the Saudi code adopts; the SBC 2024 wording, section numbers, Saudi amendments and the Civil Defense villa checklist could not be read for this study.

2 · Safe exit: the routes (lengths for information; no limit applies inside a dwelling)

From	to the stair	to the family door	Through · backup refuge
Master suite bedroom	19.5 m	37 m	suite lobby → family link → family living → gallery → landing · refuge: the suite court
Bedroom 1	14.2 m	31.6 m	family link → family living → gallery → landing · refuge: its cut-wall window
Kids' room	9.3 m	26.7 m	gallery → landing · refuge: its north window
Bedroom 2	7.8 m	25.2 m	gallery on the projection → landing · refuge: its west window
Bedroom 3	11.9 m	29.3 m	gallery on the projection → landing · refuge: its west window

Ground floor from	to the exit	Through
Maid room	9.9 m	laundry → kitchen → service door · refuge: her window
Living	18.1 m	entrance arch → riwaq → family door
Men's majlis	12.9 m	foyer → guest door

The stair: two flights 1.21 m clear, risers 0.164, goings 0.28, headroom 2.14 m, handrails both sides; 7.9 m walked on it; since 1 Nov 2026 it stops at the first floor; the roof is reached from the external stair tower (a second way out of the first floor, § 1).

glazing (E30, integrity 30 min) and move the void's high-level smoke vent to W3's tilt lites onto the courtyard (owner decision); then the route counts (the tower's openings 1.8 m from the office window, the grill 3.4 m from the gate). The court is **not** an approved refuge.

Study checks passed (7)

5 · Verification: the escape windows (IRC R310 as the SBC 1101 basis) and what is outside each

Room	Floor	Window (plans)	Size w × h	Sill above the floor	Sashes	Net clear (one sash)	Screen	Verdict	Outside: landing, rescue access
Bedroom 1	FF	10.92 1.6 → 11.79 3.2	1.82 × 1.2	0.9 (+4.50)	2 × 0.91	0.81 × 1.1 = **0.89 m²**	none	**pass**	the lane setback strip (2.5 m to the boundary) → the lane; a ladder stands in the strip, a vehicle on the lane
Bedroom 2	FF	0 11.3 → 0 12.6	1.3 × 1.2	0.9 (+4.50)	2 × 0.65	0.55 × 1.1 = **0.6 m²**	none	**pass**	the west strip (0.785 m to the M line) + the driveway beyond it → the street; a ladder from the driveway (base ≈ 1.2 m for a 4.8 m rise), a vehicle on the driveway: confirm the driveway stays open (site plan)
Bedroom 3	FF	0 17.2 → 0 18.2	1 × 1.2	0.9 (+4.50)	1 × 1	0.9 × 1.1 = **0.99 m²**	none	**pass**	as bedroom 2
Kids' room (a bedroom later)	FF	5.3 0 → 8.7 0	3.4 × 1.5	0.9 (+4.50)	3 × 1.13	1.03 × 1.4 = **1.45 m²**	none	**pass**	the north setback (3.8–5.7 m) → the north street and the parking: ladder and vehicle
Suite bedroom	FF	13.31 15 → 13.31 17.9	2.9 × 1.5	0.9 (+4.50)	3 × 0.97	0.87 × 1.4 = **1.21 m²**	none	**pass**	the east strip (3.6–9 m) → the service yard: a ladder in the strip, a vehicle from the north (site plan)
Maid room	GF	12.17 3.9 → 12.71 4.9	1.14 × 1	1.1 (+1.10)	1 × 1.14	1.04 × 0.9 = **0.93 m²**	none	**pass**	the lane setback strip at ground level, 2.5 m wide, open at both ends (to the north front and the east yard): the sill 1.40 m above the outside ground, no trapped corner

How the net clear opening is counted

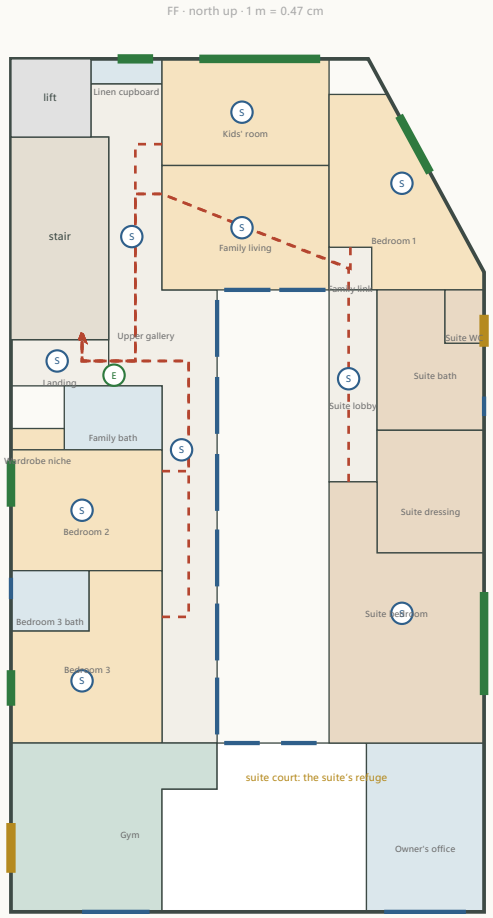
- The window line of the plans is the structural opening; black steel side-hung casements of ≤ 1.2 m, a 50 mm frame and sash edge all round, **one sash opening fully with one movement** (a single handle, no key, no restrictor). Net clear width = sash width − 2 × 0.05 height = (head − sill) − 2 × 0.05. Tilt-only, top-hung or sliding sashes do not count. **Engineer / window supplier to confirm** the sash sizes and the hardware on the shop drawings (**not verified** until then).
- The heights come from `3d/params.yaml` (the maid window listed at sill +1.10; the bedroom windows at the default sill 0.90 / head 2.10). The bedroom windows in the plans are 1.0–3.6 m wide: every one passes with one sash of ≤ 1.0 m, the maid window with its single 1.14 m sash.
- **Child restrictors** (note only): if fitted after occupation, with an adult release that needs no key and does not reduce the opening when released.

Rescue from outside: notes for the site plan

- **North** (kids' room): the setback is 3.8–5.7 m wide and opens onto the north street and the parking: a ladder or a vehicle reaches the facade. **OK**
- **Cut wall / lane** (bedroom 1, the maid room): the strip to the lane boundary is 2.5 m: a ladder (base ≈ 1.2 m for the 4.8 m rise) stands in it; a vehicle on the lane outside the wall. **OK**
- **West** (bedrooms 2 and 3): the strip inside the M line is 0.785 m: a ladder must stand on the driveway beyond the M line (3.95 m wide in `base.typ`). **not verified**: the site plan must confirm the driveway stays open and reachable at night (not fenced, not parked over); if it cannot, a ladder from the north corner reaches bedroom 2 only. The first-floor sills are 4.8 m above the ground outside (−0.30).
- **East** (the suite): the strip is 3.6–9 m wide: a ladder in the strip; the vehicle access to the east yard from the north (site plan). **OK**
- **The suite court** is the suite's fresh-air refuge, open to the sky, walled: no way down, but air and a place to call from; a rescue ladder cannot reach it (inside the roof). **OK** as a refuge, not as an exit.
- **The courtyard** (the gallery's W1 casements and the family living's W2) is likewise a refuge for whoever is in the gallery, not an exit.

5 · Verification: the alarms and the equipment on both floors (positions from design_gf.typ / design_ff.typ)





Legend Circles: **S** smoke alarm (photoelectric, mains + sealed battery, interconnected), **H** heat alarm, **CO** carbon-monoxide alarm, **G** gas detector (the shut-off valve at the cylinder cabinet), **E** extinguisher, **B** fire blanket. Red dashed: the routes; green thick: escape windows (pass); blue: other windows; green / amber on the outline: the exits.

The alarm layout, checked

Floor	Alarm	to the hob	to a shower door
GF	S · stair foot	10.9 m	8.4 m
GF	S · void ceiling	8.5 m	14.1 m
GF	S · dining	5.8 m	13.3 m
GF	S · maid room	5.8 m	2.4 m
FF	S · bed 1	—	3.7 m
FF	S · bed 2	—	2.8 m
FF	S · bed 3	—	2 m
FF	S · kids	—	6.8 m
FF	S · suite	—	8.5 m
FF	S · suite lobby	—	2 m
FF	S · gallery	—	1.3 m
FF	S · gallery N	—	4.9 m
FF	S · family living	—	4.5 m
FF	S · landing	—	3.2 m

- **Types and positions:** photoelectric alarms (fewer kitchen false alarms; ≥ 3 m from the hob: the nearest is the dining alarm at 5.8 m **OK**); ≥ 0.9 m from the shower doors (nearest 1.3 m **OK**); ceiling-mounted ≥ 0.1 m from the walls (a specification; the gypsum ceilings at ≈ +3.0 / +3.05). **The void:** one alarm on the void's ceiling (+6.98) over the living and one on the gallery's ceiling beside it; the void's alarm reacts later than a low ceiling's (smoke cools and stratifies in a 7 m space, NFPA 72 A.29.11.3.3), the gallery's and the stair foot's alarms are the early ones: the manufacturer's maximum mounting height for the void alarm is **not verified** (most ranges allow it; an outdoor-rated or high-ceiling model if the range has one).
- **Count:** 14 smoke + 1 heat (kitchen) + 1 CO (kitchen side, away from the hob) + the gas detector; 3 extinguishers (kitchen, stair foot, stair top) and the blanket.
- **Audibility:** an alarm inside every bedroom and the maid room **OK** interconnected, all sound together: ≥ 75 dBA at the pillow is met by the in-room alarm (85 dBA at 3 m per UL 217) — **not verified** until tested on site with the doors closed (a sound-level meter at the pillow, every bedroom, with the alarm in the gallery triggered).
- **Power:** the alarm circuit on the essentials circuit if the electrical designer agrees; the sealed battery keeps every alarm working in a cut; no panel, no internet.

5 · Verification: the kitchen, the exit doors · the proposals the verification produced

The kitchen as the room that slows smoke

- **Walls:** 200 mm block on all sides to the structural slab (the west wall to the courtyard, the east outer wall, the dining wall y 13.0 and the new north partition y 7.0, which the maid-wing decision specified as a partition on a ground beam: **it must run to the slab, not stop at the gypsum ceiling** → a note for the engineer, no change in plan). The kitchen ceiling is the ribbed slab's gypsum ceiling (+3.0–3.05) with the hood duct through the east wall.
- **Door to the dining:** a solid timber door 1.10 m (9.4–10.5, y 13.0) with seals and no vent: with the door closed at night the kitchen is a separate smoke compartment at household level; what it achieves: smoke from a cooking fire stays in the kitchen until the heat alarm has sounded and the family is on its way out; no time is claimed. **OK** (specification: solid, sealed, no vent, kept closed at night).
- **The laundry entrance:** the kitchen ↔ laundry opening (9.7–10.55, y 7.0) **has no door**: kitchen smoke goes straight into the laundry and reaches the maid room's door (1.00, hinged, 3.5 m away). **problem** → **P1 (proposal, not applied): a solid door in that opening** (0.85 m, the laundry side; the maid's route stays open from inside) — owner decision. Her second way is her escape window.
- **Hood and make-up air** (open item 8): the extract through the east wall straight out; no claim here.

The exit doors

Door	Width	Clear	Swing	Role
Family entrance (north wall)	1	0.9	inward, single leaf	egress door **OK**
Kitchen service door (east wall)	0.9	0.8	outward to the yard, single leaf	secondary exit
Guest entrance (west wall)	1.4	1.3	inward, two 0.70 leaves	secondary exit

- **Hardware:** every exit door with a thumb-turn or a night latch inside (no key to get out), the guest double door's second leaf on flush bolts that a child can reach; the roof stair room door outward on a thumb-turn; the laundry and the maid room doors never locked from outside while she is in. **OK** as a specification; **not verified** until the door schedule exists.
- **Clear widths:** the family entrance 0.90 m ≥ 0.813 (the dwelling's egress door); the service door 0.80 and the guest door's single leaf 0.60 are below the egress figure, which the code asks of one door only; both leaves of the guest door give 1.30. **OK**

Proposals from the verification (not applied; owner decision)

- **P1 · A door at the kitchen ↔ laundry entrance** (solid, 0.85 m, swinging into the laundry, never locked): closes the smoke path kitchen → laundry → maid room; effect: the laundry loses the open view into the kitchen; the maid's route to the service door stays (she opens the door).
- **P2 · The driveway for rescue:** keep the strip west of the M line open and unobstructed at night and show it on the site plan as the ladder stand for bedrooms 2 and 3; if the driveway cannot be guaranteed, consider making bedroom 2's and 3's windows reachable from the north corner (a wider strip is not available: the house is 0.785 m from the M line).
- **P3 · The void alarm's model:** choose a range whose specification allows the 7 m mounting height (or a high-ceiling model of the same range); otherwise rely on the gallery and stair alarms and omit the void alarm.

Verification results in one list

Point	Result	Evidence
Alarm layout: every bedroom, the gallery, the family living, the stair foot and top, the living, the void, the kitchen heat alarm	**OK**	14 smoke + 1 heat on the plans; in-room alarms in all 6 sleeping rooms; hob ≥ 3 m; shower doors ≥ 0.9 m (table, page 3)
Alarm types and mounting (wall gap, the void's height)	**not verified**	a specification; the void alarm's height per the manufacturer (P3)
Audibility in every bedroom with the door closed	**not verified**	an alarm inside each room (method); test at the pillow on site
Kitchen door and walls	**OK**	solid sealed door to the dining; block walls to the slab (note: the y 7.0 partition to the slab)
Kitchen ↔ laundry opening	**problem**	no door: smoke path to the maid room → P1
Maid window	**OK**	net 0.93 m², one casement, sill +1.10, no screen, the strip outside 2.5 m and open at both ends
First-floor bedroom windows	**OK**	all pass (page 2); rescue access north, lane and east OK; west via the driveway: **not verified** → P2
Exit doors: thumb-turns, swing, clear widths	**OK**	family door 0.90 clear; thumb-turns a specification (**not verified** until the schedule)

6 · The family escape plan · Taif / Palm House

When an alarm sounds

1. **Everyone out by the stair** and through the nearest door: the family entrance (north), the kitchen service door (east), or the guest entrance through the foyer (west). Close the doors behind you. Do not use the lift. Do not stop for anything.
2. **If the stair or the gallery is full of smoke:** stay in your room with the door closed, open your window wide for air, call 998 (Civil Defense) and the family, and show yourself at the window. **The master suite** goes out to the suite court (open to the sky) and waits there. The rescue comes from outside: the north street side, the lane side, the east yard, the driveway on the west.
3. **Meet at the assembly point:** the north parking by the gate (proposed; set with the site plan). Count everyone. Nobody goes back in.

The routes

From	Main route	Backup refuge
Master suite	suite lobby → family link → family living → gallery → stair → family door	the suite court (open to the sky); the east window
Bedroom 1	family link → family living → gallery → stair → family door	the cut-wall window (lane side)
Kids' room	gallery → stair → family door	the north window
Bedroom 2 / bedroom 3	gallery on the projection → stair → family door	the west window (driveway side)
Maid room	laundry → kitchen → service door (east yard)	her window (lane side), 1.10 m sill
Guests in the majlis	foyer → guest door (west)	—

Household rules

- **Test the alarms monthly** (the test button on one alarm sounds them all); clean them as the manufacturer says; replace every alarm after about 10 years (the date is printed on it).
- **Keep the kitchen door closed at night**, and the bedroom doors too: a closed door keeps smoke out of a room for the minutes that matter.
- **Practise the plan with the children** twice a year: the alarm sound, the two ways out of each room, the window opening, the meeting point. The children must know how to open their window and that they never hide.
- **Gas:** the cylinders stay outside in their cabinet; the shut-off valve closes by itself when the detector sounds, and by hand at the valve; the hob is never left on unattended.
- **The kitchen:** the fire blanket and the extinguisher by the service door; a pan fire is covered, never carried, never watered.
- **The roof:** the stair room door opens from inside; the roof is a way up and out of the smoke if the way down is cut, and the fire brigade can reach it from the street side.
- **Phone numbers** at the family entrance and in the kitchen: Civil Defense 998, ambulance 997, police 999 (Saudi Arabia).

Drawn from `studies/fire_escape.typ` §5; the assembly point and the rescue access are confirmed with the site plan; nothing here replaces the Civil Defense's requirements for the permit.

The second way out (south wing v2): the doors and gates on the route house → court → office → stair → outside

Door / gate	opens	lock on the way out
gallery end → gym (sliding)	sliding	no lock
gym → court (hinged)	into the court = the escape direction	free from the gym side, no key
suite bedroom → court (sliding glass)	sliding	free from the bedroom side
court → office (hinged)	into the office = the escape direction	free from the court side; key only from the office side
office → stair landing (visitors' door)	out onto the landing = the escape direction	thumb-turn inside
the gate above the office landing	not on the way down	locked (roof access only)
stair foot gate	outwards = the escape direction	panic bar; key / code outside

Verdict: **not counted as designed today**: the clerestory windows (+5.00 … +6.00) are openable ordinary glass directly over the court's movement strip; counted once they are fixed E30 glazing (the void's smoke vent moved to W3's tilt lites). The court is a route segment, not an approved refuge.